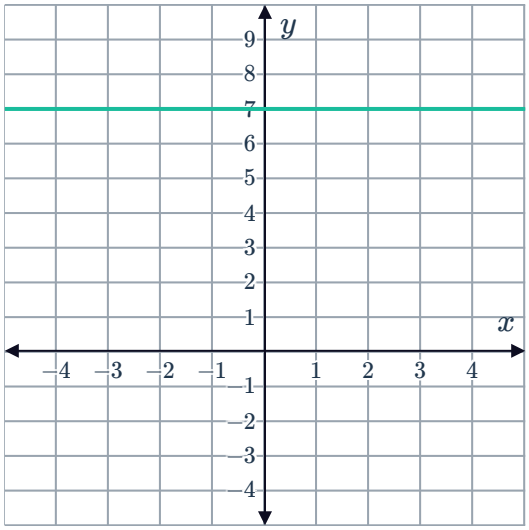
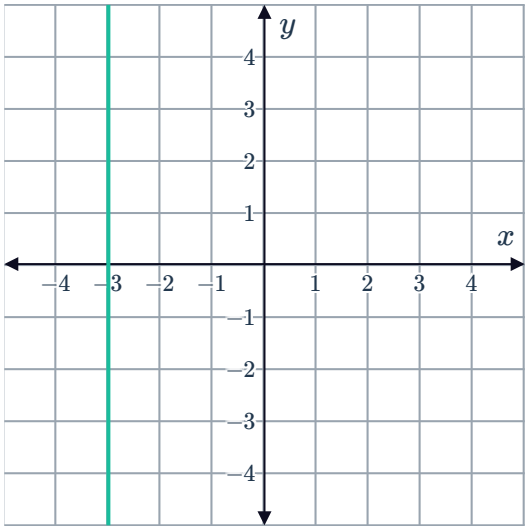
Gradient of a line

1 Describe the gradient of the lines in the following graphs:

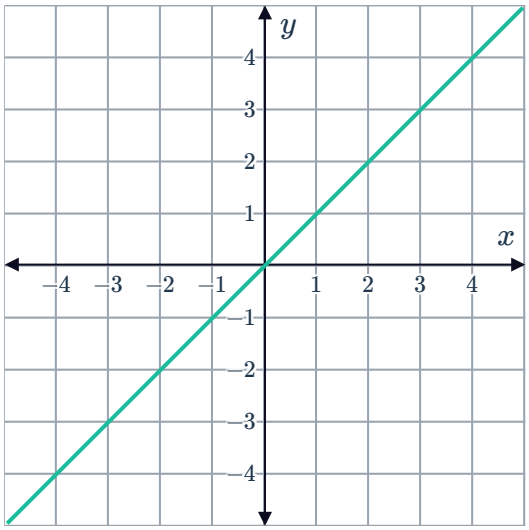
a



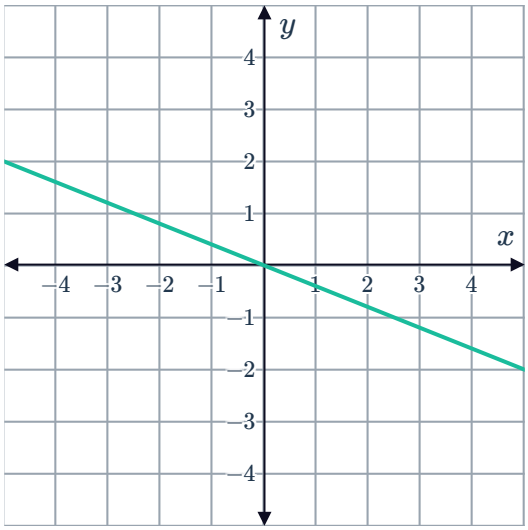
b



c



d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

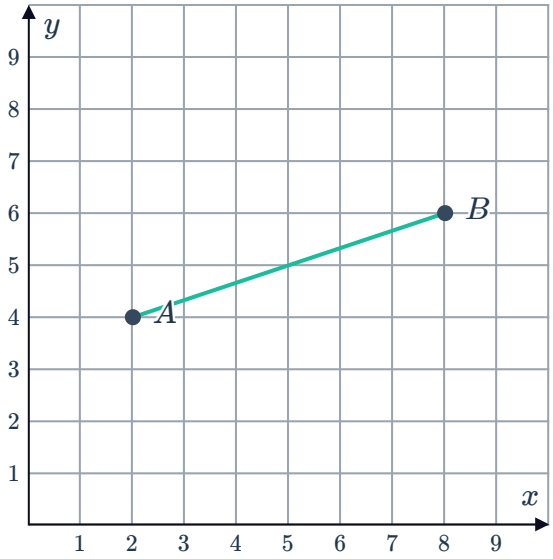
2 For each of the following intervals between points A and B:

i Find the rise.

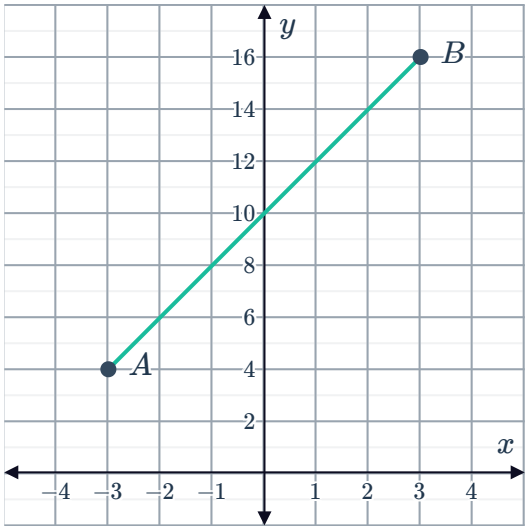
ii Find the run.

iii Find the gradient.

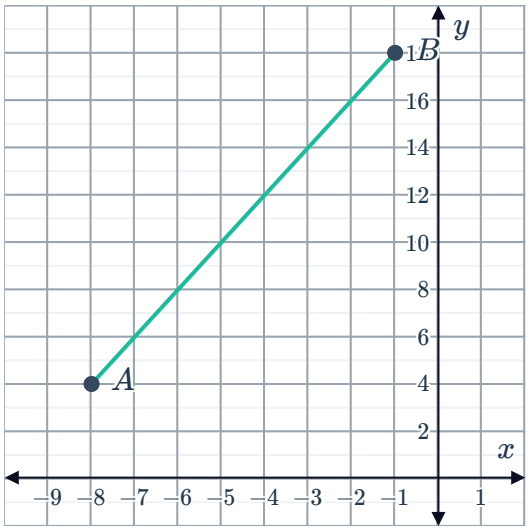
a $A\,(2,4)$ and $B\,(8,6)$



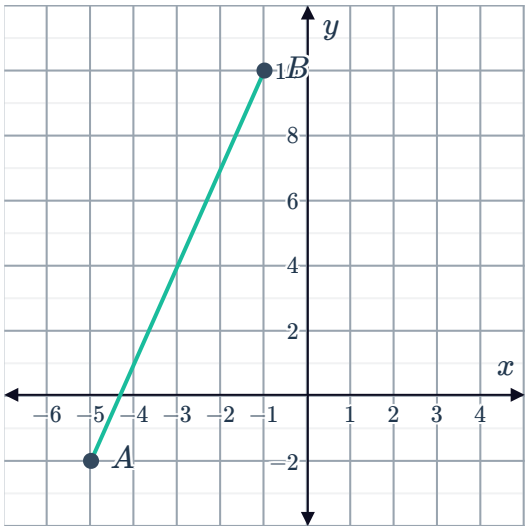
b $A\,(-3,4)$ and $B\,(3,16)$



c $A\,(-8,4)$ and $B\,(-1,18)$

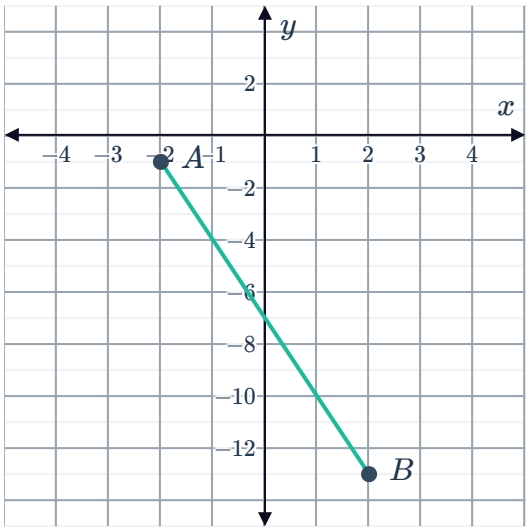


d $A\,(-5,-2)$ and $B\,(-1,10)$

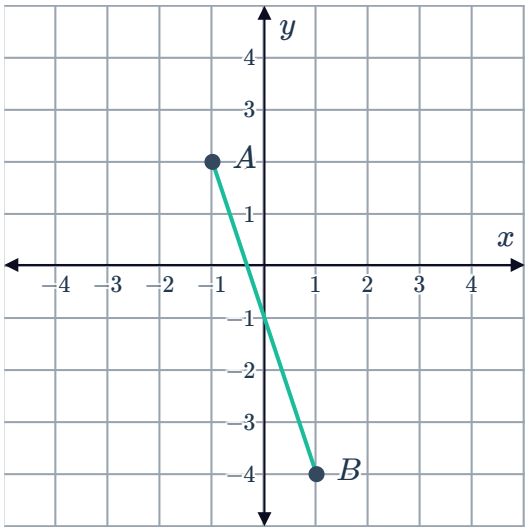


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

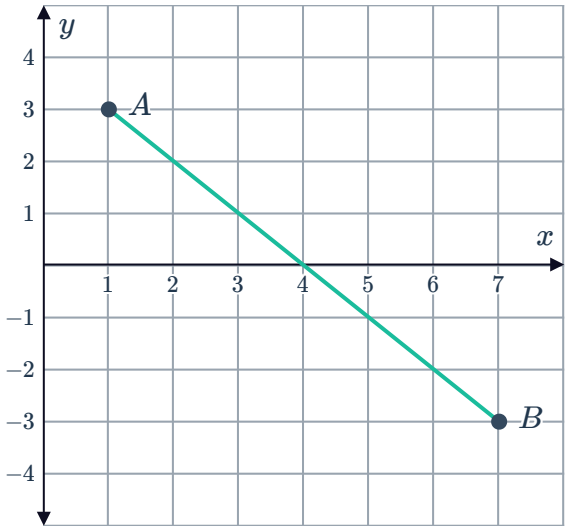
e $A\,(-2,-1)$ and $B\,(2,-13)$



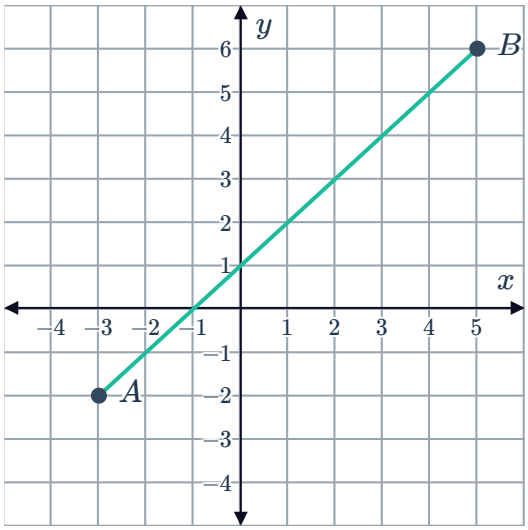
f $A\,(-1,2)$ and $B\,(1,-4)$



g $A\,(1,3)$ and $B\,(7,-3)$



h $A\,(-3,-2)$ and $B\,(5,6)$



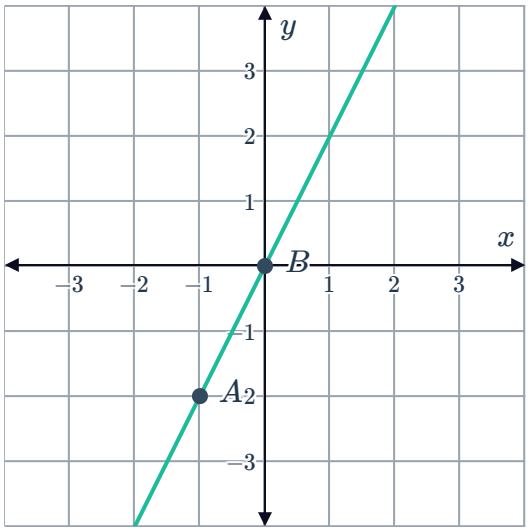
3 Explain why a vertical line has an undefined gradient.

4 State the gradient of any line parallel to the x-axis.

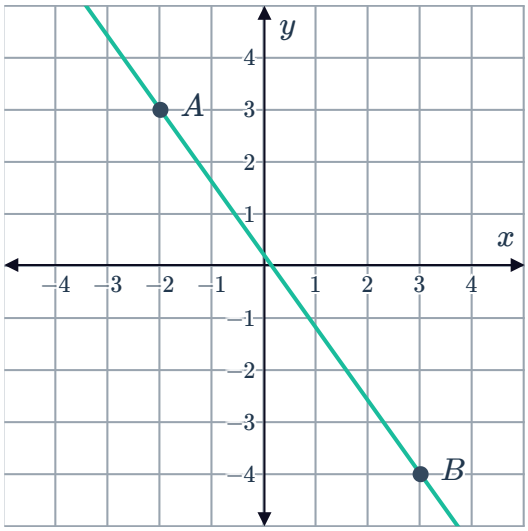
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

5 Find the gradient of the following lines passing through points A and B:

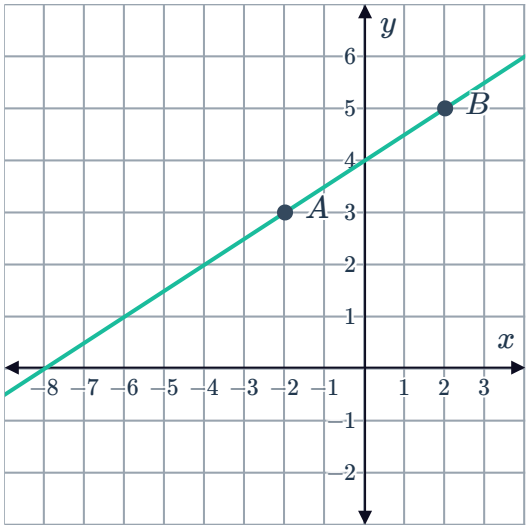
a



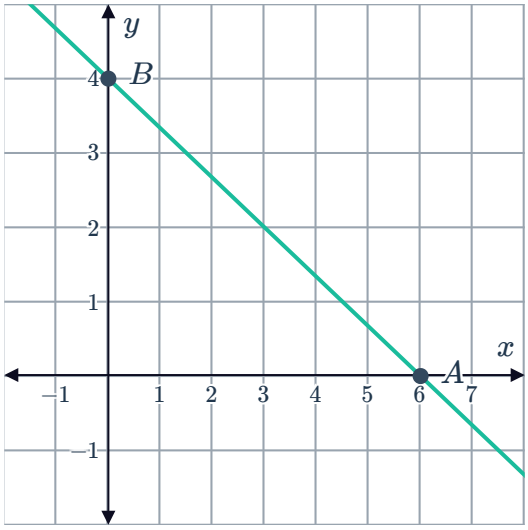
b



c



d



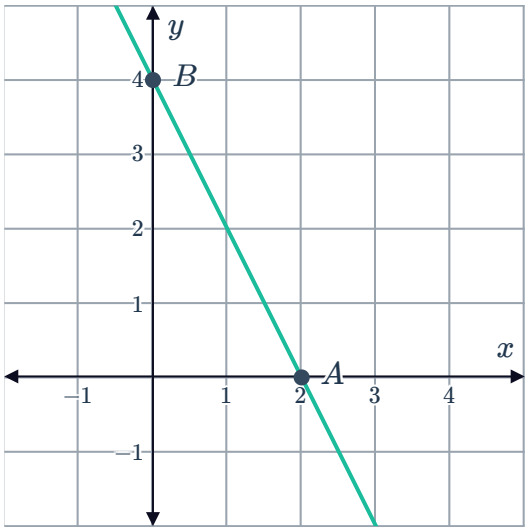
Gradient formula (Extended)

6 Find the gradient of the interval joining $A\,(-9, 4)$ and $B\,(-3, -5)$.

7 If we have two points and the slope formula $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Consider the line plotted, where $A\,(2,0)$ and $B\,(0,4)$ both lie on the line.



a Solve for the gradient of the line.

b As x increases, what happens to the value of y?

9 Find the gradient of the line that passes through the given points:

a $(-1,0)$ and $(0,3)$

b $(-4,7)$ and $(1,10)$

c $(1,-4)$ and the origin

d $(2,-6)$ and the origin

e $(6,4)$ and $(3,4)$

f $(-6,5)$ and $(4,5)$

g $(-2,-5)$ and $(-9,-12)$

h $(-3,-1)$ and $(-5,1)$

10 Consider the points $A\,(-11,-9)\,, B\,(-5,1)$ and $C\,(-2,6)$:

a Find the gradient of AB.

b Find the gradient of BC.

c Do the points A, B and C lie in a straight line?

11 Consider the following points: $A\,(26, m-24)$, $B\,(-1,m)$ and $C\,(-10,9)$.

Find m, given that A, B and C are collinear.

12 Given the gradient of the line passing through the two points, find the value of the pronumeral:

a $(4,-3)$ and $(1,t)$, gradient $=-2$

b $(5,3)$ and $(2,t)$, gradient $=-4$

c $(5,3)$ and $(d,63)$, gradient $=4$

d $(11,c)$ and $(-20,16)$, gradient $=-\dfrac{4}{7}$

13 The line $x=4$ intersects the line $y=2x-10$ at the point Q.

The line $x=-4$ intersects the line $y=2x-10$ at point R.

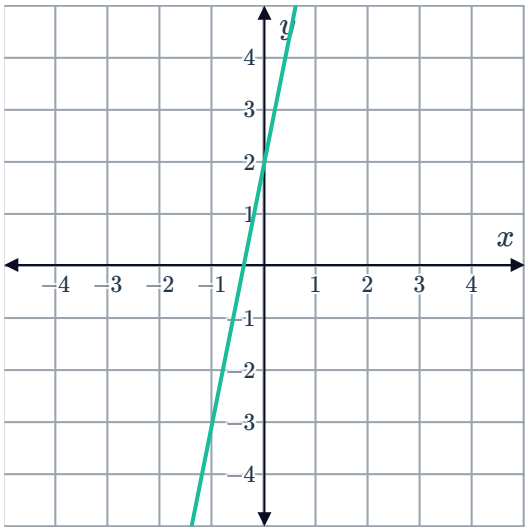
a Find the coordinates of Q.

b Find the coordinates of R.

c Find the gradient of a line that passes through Q and R.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 Consider the line $y = 5x + 2$ graphed:



a Find the y-value of the point on the line where $x = 5$.

b Find the gradient of the line.

15 Consider the line $y = -2x + 8$.

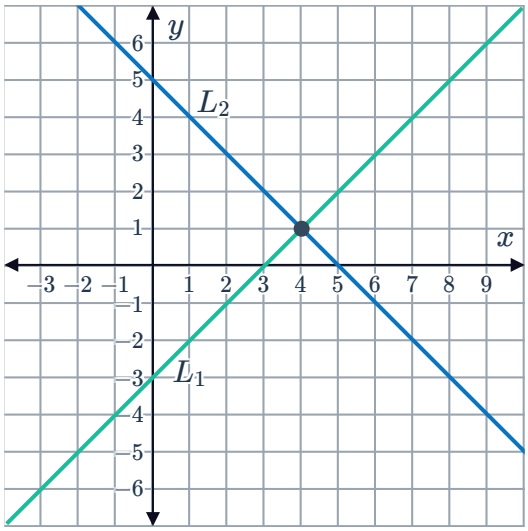
a Find:

i y-intercept **ii** x-intercept

b Sketch the line on a number plane.

c Hence, find the gradient of the line.

16 Two lines L_1 and L_2 have equations $y = x - 3$ and $y = -x + 5$ respectively. The lines and their point of intersection have been graphed:



a When $x = 4$, find the y-coordinate of the corresponding point on L_1.

b When $x = 4$, find the y-coordinate of the corresponding point on L_2.

c Find the gradient of the two lines:

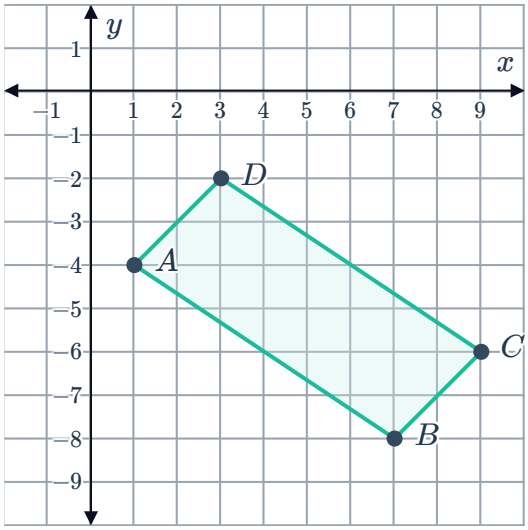
i L_1 **ii** L_2

d Find the product of their gradients.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Properties of polygons (Extended)

17 Consider the quadrilateral $ABCD$ that has been graphed on the number plane:



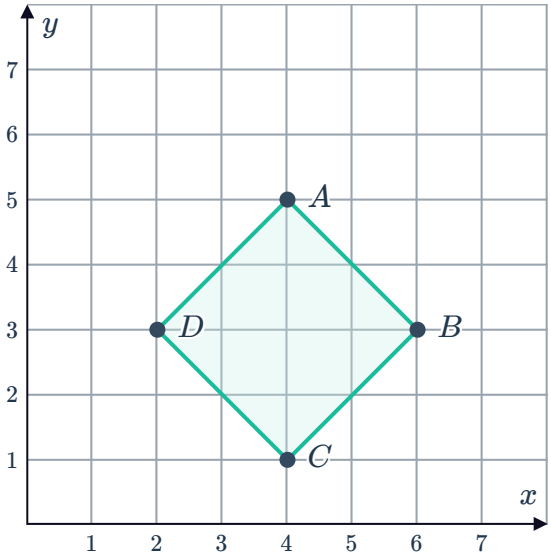
a Find the gradient of following sides:

i AB ii CD

iii AD iv BC

b What type of quadrilateral is $ABCD$? Explain your answer.

18 The 4 vertices of square $ABCD$ have been plotted on a number plane.



a Find the gradient of side AB.

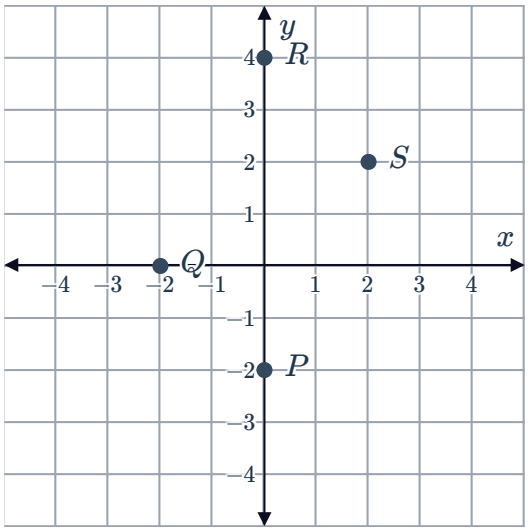
b Find the gradient of side BC.

c Find the product of the gradients in parts (a) and (b).

d If two lines are perpendicular their gradients multiply to -1. Are sides AB and BC perpendicular?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 The points $P\,(0,-2)$, $Q\,(-2,0)$, $R\,(0,4)$ and $S\,(2,2)$ are graphed below:



- **a** Find the gradient of PQ.
- **b** Find the gradient of RS.
- **c** Are PQ and RS parallel?
- **d** Are QR and PS parallel?
- **e** Identify the type of quadrilateral $PQRS$ is.

20 The vertices of $\triangle ABC$ are $A\,(9,-12)$, $B\,(4,4)$ and $C\,(-8,-5)$. The sides AB and AC have midpoints D and E respectively.

- **a** Find the coordinates of points D and E.
- **b** Find the gradient of side BC.
- **c** Find the gradient of side DE.
- **d** Are BC and DE parallel to each other?

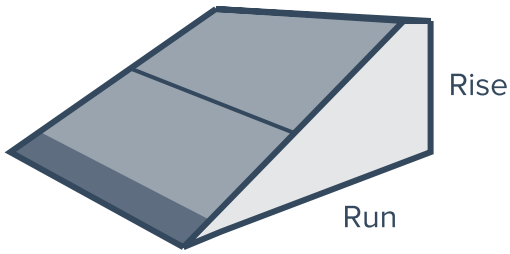
21 $A\,(-4,-2)$, $B\,(2,1)$ and $C\,(2,-4)$ are the vertices of a triangle.

- **a** Name the side of the triangle that is a vertical line.
- **b** Find the area of the triangle.

Applications

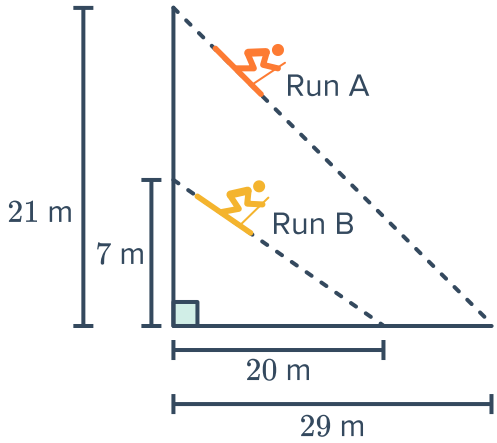
22 Consider the following ramp:

- **a** Find the gradient of this skateboard ramp if it rises 0.9 metres above the ground and runs 1 metre horizontally at the base.
- **b** The ramp can only be used as a 'beginner's ramp' if for every 1 metre horizontal run, it has a rise of at most 0.5 metres. Can it be used as a 'beginner's ramp'?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 A certain ski resort has two ski runs as shown in the diagram:



a Find the gradient of Run A. Round your answer to two decimal places.

b Find the gradient of ski run B. Round your answer to two decimal places.

c Which run is steeper?

24 A paratrooper falls to the ground along a diagonal line. His fall begins $1157\ \text{m}$ above the ground, and the line he follows has a gradient of 1.3. That is, he falls $1.3\ \text{m}$ vertically for every $1\ \text{m}$ he moves across horizontally.

How far horizontally across the ground does he land from his initial position in the sky?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647